

Physics A (H156, H556)

Exploring - Mean drift velocity ppq KSA Physics

Please note that you may see slight differences between this paper and the original.

Candidates answer on the Question paper.

OCR supplied materials:

Additional resources may be supplied with this paper.

Other materials required:

- Pencil
- Ruler (cm/mm)

Duration: Not set

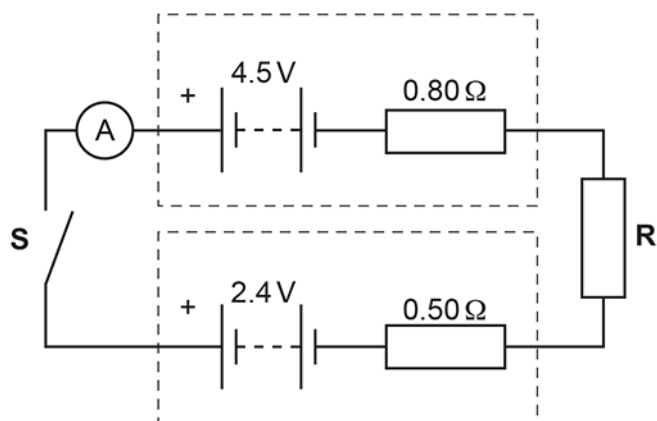
INSTRUCTIONS TO CANDIDATES

- Write your name, centre number and candidate number in the boxes above. Please write clearly and in capital letters.
- Use black ink. HB pencil may be used for graphs and diagrams only.
- Answer **all** the questions, unless your teacher tells you otherwise.
- Read each question carefully. Make sure you know what you have to do before starting your answer.
- Where space is provided below the question, please write your answer there.
- You may use additional paper, or a specific Answer sheet if one is provided, but you must clearly show your candidate number, centre number and question number(s).

INFORMATION FOR CANDIDATES

- The quality of written communication is assessed in questions marked with either a pencil or an asterisk. In History and Geography a *Quality of extended response* question is marked with an asterisk, while a pencil is used for questions in which *Spelling, punctuation and grammar and the use of specialist terminology* is assessed.
- The number of marks is given in brackets [] at the end of each question or part question.
- The total number of marks for this paper is **37**.
- The total number of marks may take into account some 'either/or' question choices.

- 1 The circuit diagram of an electrical circuit is shown below.



The positive terminals of the batteries are connected together.

One battery has electromotive force (e.m.f.) 4.5 V and internal resistance 0.80 Ω.

The other battery has e.m.f. 2.4 V and internal resistance 0.50 Ω.

R is a coil of insulated wire of resistance 1.2 Ω at room temperature.

The switch S is closed.

- i. On the diagram, draw an arrow to show the direction of the conventional current.

[1]

- ii. Calculate the current I shown by the ammeter.

$I = \dots\dots\dots$ A [3]

- iii. The insulated wire has diameter 4.6×10^{-4} m.

The number density of charge carriers in R is $4.2 \times 10^{28} \text{ m}^{-3}$

Calculate the mean drift velocity v of the charge carriers in R.

$$v = \dots\dots\dots \text{ m s}^{-1} \text{ [2]}$$

- iv. The current measured by the ammeter is smaller than that calculated in (ii). This is because the temperature of **R** increased due to heating by the current.

Without any changes to the circuit itself, state and explain what practically can be done to make the measured current the same as the calculated current..

----- [2]

- 2 Which sequence shows the materials arranged in the order of increasing number density of charge carriers?

increasing number density $\longrightarrow$

- A conductor, insulator, semiconductor
- B conductor, semiconductor, insulator
- C insulator, semiconductor, conductor
- D semiconductor, insulator, conductor

Your answer

[1]

3 Wires **P** and **Q**, made from the same metal, are connected in **parallel** across a cell of negligible internal resistance.

The table shows some data.

Wire	Length of wire	Diameter of wire	Mean drift velocity of electrons in the wire / mm s ⁻¹
P	L	d	0.60
Q	$3L$	$2d$	v

What is the mean drift velocity v of the electrons in wire **Q**?

- A 0.15 mm s⁻¹
- B 0.20 mm s⁻¹
- C 0.30 mm s⁻¹
- C 0.60 mm s⁻¹

Your answer

[1]

4

Fig. 16.1 shows the I - V characteristics of two electrical components **L** and **R**.

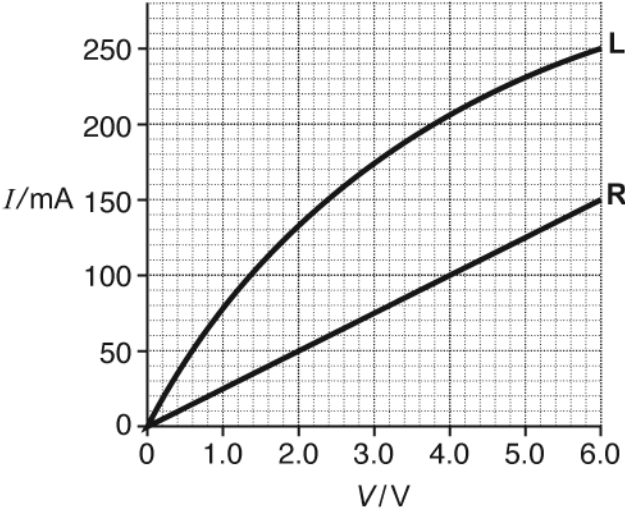


Fig. 16.1

The component **L** is a filament lamp and the component **R** is a resistor.

- i. Show that the resistance of **R** is 40 Ω .

- ii. Fig. 16.2 shows the components **L** and **R** connected in series to a battery of e.m.f. 6.0 V.

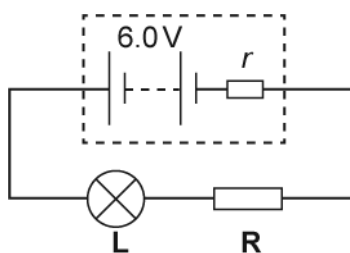


Fig. 16.2

The resistor **R** is a cylindrical rod of length 8.0 mm and cross-sectional area $2.4 \times 10^{-6} \text{ m}^2$. The current in the circuit is 100 mA.

- 1 Use Fig. 16.1 to determine the internal resistance r of the battery.

$r = \text{-----} \Omega$ [3]

- 2 Calculate the resistivity ρ of the material of the resistor R.

$$\rho = \text{-----} \quad \Omega \text{ m [2]}$$

- 3 There are 6.5×10^{17} charge carriers within the volume of R.

Calculate the mean drift velocity v of the charge carriers within the resistor R.

$$v = \text{-----} \quad \text{m s}^{-1} \text{ [3]}$$

- 5 A filament lamp **X** is part of an electrical circuit. The circuit has a battery of electromotive force (e.m.f.) 6.0 V and negligible internal resistance. The potential difference across the lamp can be increased **continuously** from 0 to 6.0 V. This potential difference is measured using a voltmeter.

The lamp glows brightly at 6.0 V.

- i. Draw a circuit diagram for this electrical arrangement.

[2]

- ii. Describe and explain the variation of the resistance of this lamp as the potential difference across it is changed from 0 to 6.0 V.

[4]

- iii. The filament lamp **X** is now connected in a different circuit as shown in Fig. 16.

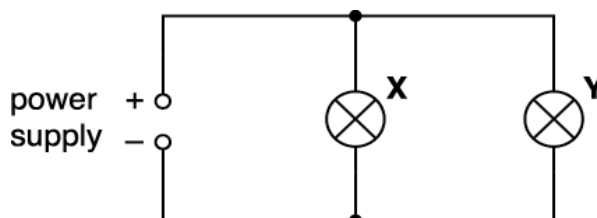


Fig. 16

The power dissipated in **X** is three times more than the power dissipated in the filament lamp **Y**. The filament wire of lamp **X** has a diameter half that of lamp **Y**.

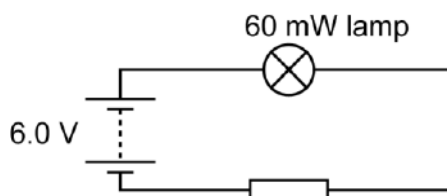
The filament wires of **X** and **Y** are made of the same material and are at the same temperature.

Calculate the ratio

$$\frac{\text{mean drift velocity of charge carriers in lamp X}}{\text{mean drift velocity of charge carriers in lamp Y}}$$

ratio = [3]

6(a) A battery is connected in series with a lamp and a resistor as shown.



The battery has e.m.f. 6.0 V and negligible internal resistance. The potential difference across the lamp is 2.4 V and it dissipates 60 mW. The resistor has cross-sectional area of 2.0 mm². The number density of charge carriers (free electrons) within the resistor is $1.4 \times 10^{25} \text{ m}^{-3}$.

Calculate the resistance R of the resistor.

$R = \dots \dots \dots \Omega$ [3]

(b) Calculate the mean drift velocity v of the free electrons within the resistor.

$v = \dots \dots \dots \text{ m s}^{-1}$ [3]

- (c) The number density of the free electrons in the connecting wires is greater than that of the resistor. The connecting wires have the same diameter as the resistor. State and explain whether the mean drift velocity of the free electrons would be smaller, the same, or larger than your value in (b).

----- [2]

- 7 A copper wire is connected across a cell. The conduction electrons within the copper wire move.

Which statement is correct about these electrons?

- A. They drift towards the negative end of the cell.
- B. They have random speeds because of collisions with other electrons.
- C. They travel through the wire at the speed of light.
- D. They collide with vibrating copper ions.

Your answer ☐

[1]

END OF QUESTION PAPER

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
1		i	Arrow in anticlockwise direction	B1	Allow this mark for correct direction shown on diagram either on or off connecting wires <u>Examiner's Comments</u> This question required the candidates to appreciate that the sum of the emfs will lead to an anticlockwise conventional current. This question was answered well by the majority of candidates, however a number put two directions on, one from each cell.  Misconception The unusual setting out of the circuit meant that some candidates were unsure whether parts of the circuit were in series or parallel. This could have been overcome by following the circuit or even by redrawing it.
		ii	$(E =) 4.5 - 2.4$ or $(R_T =) 0.80 + 0.50 + 1.2$ $4.5 - 2.4 = I \times (0.80 + 0.50 + 1.2)$ $I = 0.84 \text{ (A)}$	C1 C1 A1	$E = 2.1 \text{ (V)}$; $R_T = 2.5 \text{ (}\Omega\text{)}$ Treat missing 1.2 resistance as TE Allow 2 marks for 2.8 (A); $E = 6.9 \text{ V}$ used <u>Examiner's Comments</u> This calculation required the candidate to set out the whole circuit in one. Around one third did not score any marks on this question as they attempted to treat each cell individually and then produce some form of average. Other common misunderstandings included treating the 0.5 ohm and 0.8 ohm resistors as if they were in parallel, and adding the emfs.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		iii	$(I = Anev)$ $0.84 = \pi \times (2.3 \times 10^{-4})^2 \times 4.2 \times 10^{28} \times 1.60 \times 10^{-19} \times v$ $v = 7.5 \times 10^{-4} \text{ (m s}^{-1}\text{)}$	C1 A1	Possible ECF from (ii) Note answer is $2.5 \times 10^{-3} \text{ (m s}^{-1}\text{)}$ for $I = 2.76 \text{ (A)}$ Allow 1 mark for 1.9×10^{-4} ; diameter used as radius <u>Examiner's Comments</u> This question was well done by a large number of candidates, many of whom scored full marks by correctly following through with their value of current from the previous part. Few candidates used the diameter instead of the radius when determining the cross sectional area, and for the most part the setting out of the calculation meant that credit could be given even if arithmetic errors occurred later.
		iv	Sensible suggestion, e.g. use a water bath / fan / only switch on when taking readings Need to lower the temperature / reduce resistance of R	M1 A1	Allow keep the surroundings cold Allow to keep the temperature / resistance constant OR allow increase in temperature increases resistance <u>Examiner's Comments</u> Candidates were expected to provide any method of cooling the circuit, or preventing it heating in the first place. A wide variety of solutions were given and as long it is viable, it was credited.  Misconception Some candidates gave perfectly viable solutions, but these involved changes to the circuit, which was not allowed in the question. It is very important to make sure than any response does fit what is being asked.
			Total	8	

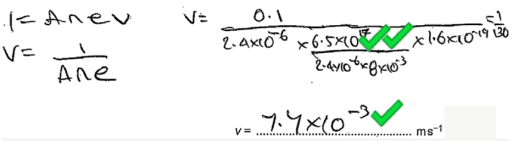
Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
2			C	1	
			Total	1	
3			B	1	Examiner's Comments This was a question on combining together three important expression in the topic of electricity; $V = IR$, $R = \rho L/A$ and $I = Anev$. On top of this, there was the additional information that P and Q were in parallel and hence the potential difference across each wire was the same. The mean drift velocity v of the electrons is given by the expression $v = \frac{V}{ne\rho L} \propto \frac{1}{L}$. The cross-sectional area A, and hence the diameter d of the wire has no effect on v. The relationship above implies that for wire Q, $v = \frac{1}{3} \times 0.60 = 0.20 \text{ mm s}^{-1}$. The correct answer is B. All the distractors were equally popular. About a third of the candidates, mostly from the very top end of the ability range, were successful in this very demanding question.
			Total	1	

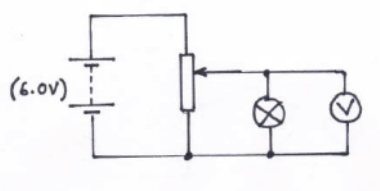
Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
4		i	$(R =) \frac{6.0}{0.150}$ $R = 40 \Omega$	M1 A0	Allow any correct value of $V (\pm 0.1 \text{ V})$ divided by the correct value of $I (\pm 10 \text{ mA})$ from the straight line for R <u>Examiner's Comments</u> The majority of the candidates scored 1 mark here for clearly using the graph to show the resistance of R to be 40Ω. Most used a data point from the straight line. A significant number also used the idea that the gradient of the straight line is equal to the inverse of the resistance. However, candidates are reminded that resistance is equal potential difference divided by current, but in this context of a straight line through the origin, determining resistance from the gradient was allowed. Of course, determining the gradient of a curve is simply incorrect physics for determining resistance.
		ii	$(V_L =) 1.4 \text{ (V)}$ or $(V_R =) 4.0 \text{ (V)}$ or $(R_T =) 6.0/0.1 \text{ (}\Omega\text{)}$ $(V_{\text{terminal}} =) 5.4 \text{ (V)}$ or $(V_r =) 0.6 \text{ (V)}$ or $(r =) 60 - 54 \text{ (}\Omega\text{)}$ $r = 6.0 \text{ (}\Omega\text{)}$	C1 C1 A1	Allow full credit for other correct methods Possible ECF from (i) Allow $\pm 0.1 \text{ V}$ for the value of p.d. from the graph Note getting to this stage will also secure the first C1 mark Allow 1 SF answer here without any SF penalt <u>Examiner's Comments</u> This was a discriminating question with many of the top-end candidates effortlessly getting the correct answer of 6.0Ω for the internal resistance r. The most common error was omitting the resistance of the filament lamp in the calculation. This gave an incorrect value of 20Ω for the internal resistance. Candidates doing this still managed to pick up 1 mark for the total resistance of 60Ω.

Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
iii	$\rho = \frac{40 \times 2.4 \times 10^{-6}}{8.0 \times 10^{-3}}$ (Any subject) $\rho = 0.012 (\Omega \text{ m})$	C1 A1	Allow ECF Allow 1 mark for either 0.018 for using 60 Ω, 0.016(2) for using 54 Ω or for 0.0018 for 6.0 Ω Examiner's Comments The success in this question depended on understanding the term n in the equation $I = Anev$ given in the Data, Formulae and Relationship booklet. A significant number of candidates took n to be the total number of charge carriers within the volume of R, instead of the number of charge carriers per unit volume (number density). Those who appreciated this had no problems coping with prefixes and powers of ten. The correct answer was $7.7 \times 10^{-3} \text{ m s}^{-1}$. Using 6.5×10^{17} for the number density, gave an answer of $4.0 \times 10^5 \text{ m s}^{-1}$; examiners credited 1 mark for this incorrect answer, mainly for the manipulating and using the equation $I = Anev$. Exemplar 6  This exemplar illustrates a perfect answer from a C-grade candidate. The equation has been rearranged correctly and the substitution is all correct and easy to follow. The number density n has not been calculated separately – it forms an integral part of the whole calculation. The one big benefit of this is that you do not end up with rounding errors. A decent technique demonstrated here. All correct for 3 marks.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		iv	$n = \frac{6.5 \times 10^{17}}{2.4 \times 10^{-6} \times 0.008}$ or $n = 3.385 \times 10^{25} \text{ (m}^{-3}\text{)}$ $v = \frac{0.100}{2.4 \times 10^{-6} \times 3.385 \times 10^{25} \times 1.60 \times 10^{-19}}$ (Any subject) $v = 7.7 \times 10^{-3} \text{ (m s}^{-3}\text{)}$	C1 C1 A1	Note do not penalise again for the same POT error Allow 1 mark for $4(.0) \times 10^5 \text{ (m s}^{-1}\text{)}$; $n = 6.5 \times 10^{17}$ used
		Total		9	
5		i	Correct circuit with a battery, potential divider, lamp and voltmeter. 	B1	
		i	Correct symbols used for all components.	B1	Allow: A cell symbol for a battery
		ii	Description: The temperature of the filament increases. (AW)	B1	
		ii	The resistance of the lamp increases	M1	
		ii	from a non-zero value of resistance.	A1	Allow 'when cold the resistance is small'
		ii	Explanation: Resistance increases because electrons/charge carriers make frequent collisions with ions. (AW)	B1	
		iii	($P = VI$) current in X is 3 times the current in Y Or area of X is 4 times smaller than area of Y	C1	Allow other correct methods.
		iii	$\frac{3}{I = Anev \text{ and ratio} = 0.25}$	C1	
		iii	ratio = 12	A1	
		Total		9	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
6	a		current = $\frac{0.060}{2.4}$ or current = 0.025 (A)	C1	Note answer to 3 sf is 144 Ω
			$R = \frac{6.0 - 2.4}{0.025}$	C1	
			$R = 140 (\Omega)$	A1	
	b		$I = Anev$ and $A = 2.0 \times 10^{-6} (\text{m}^2)$	C1	Allow any subject Possible ecf
			$0.025 = 2.0 \times 10^{-6} \times 1.4 \times 10^{25} \times 1.60 \times 10^{-19} \times v$	C1	
			$v = 5.6 \times 10^{-3} (\text{m s}^{-1})$	A1	
	c		The current is constant, therefore $v \propto n^{-1}$.	M1	
			The mean drift velocity is therefore smaller.	A1	
			Total	8	
7			D	1	
			Total	1	